

Project Design Phase-II

Data Flow Diagram & User Stories

Date	01 September 2026
Project Type	Individual Project
Project Name	Software Installation Request Automation in ServiceNow Using Service Catalog and Workflows
Maximum Marks	4 Marks

Data Flow Diagram:

A Data Flow Diagram (DFD) is a visual representation of how a software installation request moves through ServiceNow — from submission to fulfillment or exception handling. The flow below (sequential, top to bottom) traces the request lifecycle built in this project:

1. Employee (Requester) opens Service Portal / Catalog and selects 'Software Installation Request'.
2. Employee fills catalog variables: Software Name, Version Required, License Justification, Urgency — and clicks Order Now.
3. A Request (sc_request) and Requested Item (sc_req_item) are created; the linked Workflow ('Software Request Workflow') is triggered via the Process Engine.
4. Workflow runs Manager Approval (Approval - Group activity) — condition: approval = requested.
5. If Approved → Create Task activity generates a Software Fulfillment Task (sc_task) assigned to the Software Support Team.
6. If Rejected → Requester is notified and the request is closed.
7. IT Support fulfils the task and updates state (Open → Work in Progress → Completed); requester is notified.
8. If the Requested Item state changes to 'On Hold' (e.g., license shortage), a Business Rule auto-creates an Incident, assigned to Software Support, linked back to the request.
9. All configuration (catalog item, variables, workflow, business rule) is captured in an Update Set for migration between instances.

User Stories

User stories for the Software Installation Request Automation solution, mapped to each stakeholder identified in the requirement analysis.

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User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
End User (Requester)	Request Submission	USN-1	As an employee, I can submit a request for company-approved software by entering the software name, version, justification, and urgency.	I can see my request confirmed with a request number (e.g., REQ0010006).	High	Sprint-1
End User (Requester)	Status Visibility	USN-2	As an employee, I can track the status of my software request (Approved / Rejected / In Progress / Completed).	I can view request stage and delivery date from the portal.	High	Sprint-1
Approver (Manager/Security)	Approval Routing	USN-3	As a manager, I receive an approval task automatically when a request is routed to me, so I can approve or reject it.	I can approve/reject and the workflow proceeds accordingly.	High	Sprint-1
IT Support / Fulfillment Team	Task Fulfillment	USN-4	As IT support, I receive an automatically assigned fulfillment task once a request is approved, with all details pre-filled.	I can view, work on, and complete the task from my queue.	High	Sprint-1
IT Support / Fulfillment Team	Exception Handling	USN-5	As IT support, an incident is automatically raised if a software request cannot be fulfilled due to a license shortage.	An Incident record is created and assigned to Software Support.	Medium	Sprint-2
ServiceNow Admin	Configuration Migration	USN-6	As an admin, I can package all catalog/workflow customizations into an Update Set and move them across environments.	Update Set exports, previews, and commits without collisions.	Medium	Sprint-2